



Robust optimization and pricing of Peer-to-Peer energy trading considering battery storage

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ABSTRACT

Peer-to-peer energy trading is one of the greatest systems that helps to increase the use of renewable energy, such as photovoltaics and wind turbines, in households due to the escalating environmental issues. This study designs a peer-to-peer energy trading system under the uncertainty of renewable energy generation and peers' demands. This study uses a cardinality-constrained robust optimization approach to tackle uncertainties, and it is solved to minimize the overall costs of energy trading, battery depreciation, and load shedding by the Alternating Direction Method of Multipliers method. This model considers peers who are equipped with distributed generation and batteries. Moreover, this local grid has a central battery storage in order to reduce load shedding and dependence on the power grid. Furthermore, the Nash bargaining theory is modeled to calculate energy prices that are traded by peers in order to boost the benefit of all peers and encourage them to use renewable energy as well as participate in peer-to-peer energy trading. Finally, based on data from Iran, the mathematical model is solved to show the effectiveness of using peer-to-peer energy trading that not only can save significant power but also make substantial profits for peers.

1. Introduction

The worldwide community has given climate change substantial consideration over the past few decades (Persson & Dzebo, 2019), and global warming has been one of the most important issues for various countries due to its adverse consequences (Karmalkar and Bradley, 2017; Seif et al., 2023). It is worth noting that 73.2% of greenhouse gas (GHG) emissions are attributed to energy use, which means that non-renewable resources are a major contributor to global warming (Ritchie and Roser). In contrast, using renewable sources results in less carbon dioxide emissions, which in turn can be used as an effective strategy to mitigate climate change (Lin & Zhu, 2019). Hence, using renewable energy is necessary to mitigate GHG emissions and climate change (Levenda et al., 2021; Ghadami et al., 2021). Global energy consumption increased by 21% in 2017 compared to the previous year, and the U.S. Energy Information Administration predicts that energy consumption will be projected to grow by approximately 50% in Asia by 2050. The world is turning to renewable energy systems (RESs) since it is uncertain that traditional energy systems will be able to meet this expanding demand (Ali et al., 2020). In every country, energy serves as the cornerstone for other economic factors and is crucial for addressing

societal expectations. Renewable energy is promising clean energy from natural resources such as sunlight, wind, earth, and plants (Bull, 2001).

Energy is considered an essential input for economic activities (Sahebi & Nickel, 2014). Distributed generation (DG) based on rooftop solar photovoltaic (PV) systems with energy storage systems is an alternative energy generation technology to reduce GHG emissions (Nguyen et al., 2018). The global energy market will experience fundamental shifts by growing consumer demand for renewable sources and storage systems (Council, 2016). The widespread use of DG technologies such as solar PV has turned passive consumers into active producers (Basnet & Zhong, 2020). Producers can store their surplus energy in batteries or sell it to others who suffer from energy shortages (Zhang et al., 2016). Peer-to-peer (P2P) energy sharing has become one of the most promising approaches to coordinating residential microgrids (Li & Zhang, 2021). Studies show that P2P grids with battery energy storage systems (BESS) can provide significant savings in energy (Nguyen et al., 2018). Sharing energy increases prosumers' flexibility, both for selling electricity to peers in need and purchasing electricity from other peers who have surplus energy (Weber, 2015). The development of energy models for the application of renewable energy and batteries, as well as new energy market structures has always been

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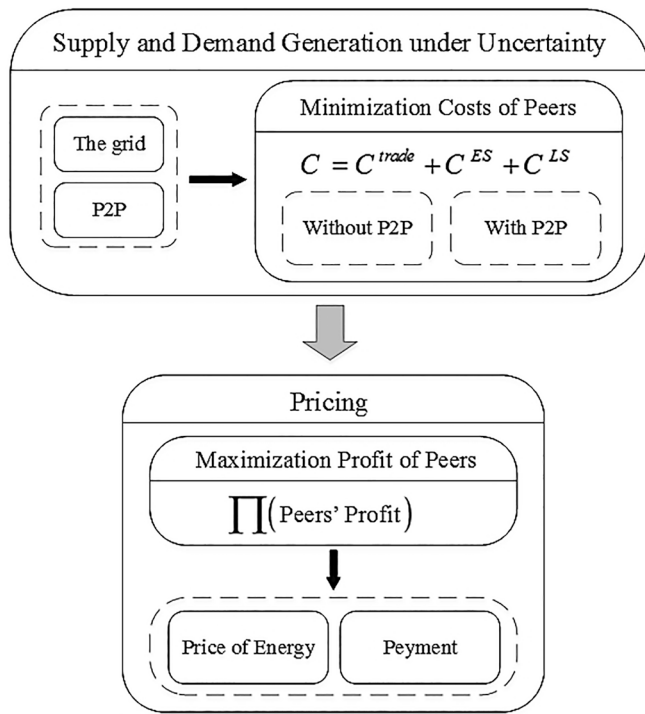


Fig. 1. P2P energy trading method to optimization costs and price.

considered to maximize earnings (Ackermann et al., 2001; Delavar and Sahebi, 2020).

Feed-in tariff (FiT) initiatives have been introduced in many countries to encourage the utilization of RESs and reap its benefits, in which governments buy surplus electricity from producers. Some prosumers also store their surplus energy for later self-consumption at peak loads (Long et al., 2017). A major challenge has arisen in encouraging producers to trade energy with each other. An appropriate energy trading model focuses on both effective management of energy transmission and pricing policies, which makes peers interested in engaging in P2P trading, whether as electricity buyers or sellers (Liu, Wu, & Li, 2019). The successful establishment of a P2P energy trading encounters two major challenges: (i) how to optimally plan energy transition, and (ii) how to design appropriate financial incentives (Saad et al., 2016). The P2P model for energy trading and energy pricing has been proposed to address these challenges (Li, Ye, & Strbac, 2020). Since facilitating energy sharing is difficult without a pricing mechanism, internal pricing mechanisms are an important aspect of implementing P2P energy sharing (Tushar et al., 2018). In some cases, a third factor is introduced as a supervisor in P2P in order to determine energy price, but in other cases, the pricing mechanism is done directly among peers (Newlands, Lutz, & Fieseler, 2018).

In this study, as is shown in Fig. 1, we design P2P energy trading that peers are equipped with different DG and battery storage, taking into account the uncertainty in the amount of energy generation as well as the demand of peers, to obtain the amount of power that is traded between the grid, peers, and battery storage.

Based on the amount of energy trading among peers, the price of power in P2P trading is calculated in order to maximize the profits of all peers. The robust method is used to deal with uncertainty, and Nash bargaining theory is also used for pricing. In this P2P grid, we consider central battery storage that is charged by renewable energy from peers in order to reduce the load shedding and also dependence on the grid. In addition, this study compares a local grid without and with P2P trading and elucidates which one has more benefit for peers. In this P2P, no third member is present as an observer, and the P2P energy trade is conducted on the blockchain platform. Overall, some innovations are as follows:

- Considering different DG and battery storage for peers, and central battery storage for the local grid in order to reduce load shedding.
- Considering uncertainty for energy generation and the demand of peers, and tackling it with a cardinality-constrained robust optimization approach; the proposed model is solved to minimize the overall costs of energy trading.
- The Nash bargaining theory is employed to calculate energy prices in order to improve peers' profit and encourage them to use renewable energy.
- Drawing a comparison between energy trading with considering P2P and without it.
- Considering the effect of time on the generation power of DG and the price of power. Moreover, energy trades are made on the blockchain directly among peers without a supervisor.

The rest of this paper will address the issues as follows. Section 2 reviews the literature on energy systems, P2P, and energy trading. Afterward, Section 3 depicts the problem description, and Section 4 presents the mathematical model considering uncertainty and robust optimization. After solving the P2P energy trade model, the pricing model is presented in Section 5. Section 6 analysis the mathematical model through a case study and describes the results. In Section 7, sensitive analyses of the model are conducted. Finally, Section 8 describes the conclusion.

2. Literature review

The complexity of electricity distribution networks as a result of various energy generation and storage technologies necessitates an urgent need to alter the organization and management of both energy generation and distribution. P2P energy sharing is one of the most promising approaches to coordinating domestic microgrids to form sustainable communities. Tushar et al. (Tushar et al., 2019) offered how a motivational psychology framework could be applied to design a P2P energy trading to increase user engagement. Sæther et al. (Sæther, del Granado, & Zaferanlouei, 2021) analyzed both P2P power trading and local grid flexibility. They also assessed how a local market influences peak load management of electricity demand. For this analysis, a linear programming (LP) model was developed, which showed and simulated the characteristics of the local electricity grid. Nguyen et al. (Nguyen et al., 2018) proposed an optimization model for maximizing economic benefits in a P2P energy trading system; the aim was to investigate the feasibility of the energy storage system in the P2P grid with the study of financial earnings.

Huang et al. (Huang et al., 2020) developed a P2P energy trading model in microgrids with solar PV and BESS. He et al. (He, Liu, & Zhang, 2021) offered a P2P energy trading framework that allowed solar PV prosumers and consumers to engage in a community co-market established by an energy pawn. The energy pawn was responsible for setting up, connecting, managing, and maintaining the P2P grid and had a publicly available BESS that could be utilized to facilitate energy sharing in the community. Alam et al. (Alam, St-Hilaire, & Kunz, 2017) developed a mixed-integer nonlinear programming (MINLP) model to integrate demand-side management with P2P energy trading for smart homes. However, no trading mechanism was considered in that study, and the simulation was performed only for two households, which was not enough to show the earnings from P2P trading. Elkazaz et al. (Elkazaz, Sumner, & Thomas, 2021) developed a novel method called the 'Hierarchical Peer-to-Peer Energy Management method', which effectively reduced household energy costs. This hierarchical system consisted of three levels: (i) selection level, (ii) P2P grid management level, and (iii) home management level. First, the daily energy cost of each household was optimized individually at the lower layer; then the results were optimized through an algorithm, in which peers were selected such that they could have a better reduction in operating costs.

Energy generation from renewable sources depends on

Table 1
Review of previous literature and research.

Reference	Year	Model	Objective function		Solving approach		Consideration					Purpose of problem
			S	M	E	S	B	U	C	G	P	
Liu et al. (Liu et al., 2015)	2015	NLP	●	–	●	–	–	–	–	●	–	Cooperative microgrids with P2P
Rosell et al. (Olivella-Rosell et al., 2016)	2016	MIP	–	●	–	●	–	–	●	●	–	Designing micro-market for DG
Tushar et al. (Tushar et al., 2016)	2016	Game Theory	●	–	●	–	●	–	–	–	●	Energy storage sharing in smart grid
Hermana et al. (Alvoro-Hermana et al., 2016)	2016	NLP	●	–	●	–	–	–	●	–	–	P2P and electric vehicles
Long et al. (Long et al., 2017)	2017	NLP	●	–	–	●	–	–	–	●	–	P2P in low voltage electrical distribution grids
Sorin et al. (Sorin et al., 2017)	2017	LP	●	–	–	●	–	–	●	–	–	P2P impacts of a consumer-oriented grid
Moret et al. (Moret & Pinson, 2018)	2018	NLP	●	–	–	●	–	–	●	●	–	P2P and community-based markets
Morstyn et al. (Morstyn, Teytelboym, & McCulloch, 2018)	2018	Game Theory	–	●	–	●	–	–	–	●	–	Bilateral contract for P2P
Lüth et al. (Lüth et al., 2018)	2018	LP	●	–	●	–	●	–	●	–	–	The role of batteries in P2P
Zepter et al. (Zepter et al., 2019)	2019	SLP	–	●	●	–	●	●	–	–	–	P2P trading with the day-ahead and intraday markets
Alam et al. (Alam, St-Hilaire, & Kunz, 2019)	2019	MILP	●	–	●	–	–	–	●	–	–	P2P among smart homes
Guo et al. (Guo et al., 2020)	2020	MINLP	●	–	–	●	–	●	●	–	–	Considering P2P network uncertainty
Hua et al. (Hua et al., 2020)	2020	MINLP	–	●	–	●	–	–	●	●	–	A blockchain-based P2P and carbon markets
Kanakadhurga et al. (Kanakadhurga & Prabakaran, 2021)	2021	NLP	●	–	–	●	–	–	●	●	–	Demand response-based P2P energy trading
Chang et al. (Chang, Xu, & Sun, 2022)	2022	MIP	–	●	●	–	–	●	●	●	–	Vertex scenario-based robust P2P energy trading
Xia et al. (Xia et al., 2022)	2022	MILP	●	–	●	–	–	●	–	●	●	P2P energy trading under uncertainty of DG
This study	–	MILP	–	●	●	–	●	●	●	●	●	P2P with DGs and batteries under uncertainty of energy generations and demands and then energy pricing

environmental conditions such as climate fluctuations which cause uncertainties. Gernaat et al. (Gernaat et al., 2021) used climate and integrated assessment models to estimate the impact of environmental conditions on renewable energy. López-García et al. (López-García, Torreglosa, & Vera, 2019) offered a strategy for the energy imbalance problem in P2P energy trading grids based on dividing the energy transaction into energy fragments. To that end, a novel control initiative was proposed in which suppliers and customers use Information and Communications Technology (ICT) to buy and sell the agreed amount of energy.

Pricing energy in the P2P energy market has become more challenging due to the growing energy demand and the integrated renewable energy sources. Participants require a fair mechanism that ensures the basic goals of energy, i.e., the saving of consumers' energy bills and the profitability of the seller. To achieve these goals, Z. Li and Ma (Li & Ma, 2020) attempted to achieve a better understanding of P2P electricity trading in the community, including the impact of trading strategies, factors determining the amount and price of transactions, and other factors affecting the improvement of energy transactions in the community. Weber et al. (Weber, 2016) developed an overlapping model for pricing analysis and consumer selection with and without a common market. This model quantitatively analyzed the impacts of P2P economics on consumer demand, energy prices, consumer surplus, profit, and social welfare. Tang et al. (Tang, Kim, & Wang, 2019) investigated the price-determining factors for P2P facilities in their research.

PankiRaj et al. (PankiRaj, Yassine, & Choudhury, 2019) provided an optimization model using an auction-based mechanism for P2P energy trading in a smart grid in which appropriated infrastructure and communications were considered to allow users to trade energy on a P2P foundation. This mechanism required a full understanding of the various constraints associated with grid infrastructure, e.g., the status of the connecting routes of suppliers and consumers. Amin et al. (Amin et al., 2020) provided a uniform pricing initiative for energy trading with an energy allocation policy for the P2P energy trading market. According to consumer preferences, a new method was offered for pricing energy trade, which ensured increased saving for consumers and profitability of sellers.

Vieira et al. (Vieira & Zhang, 2021) studied the application of

blockchain technologies and auction mechanisms to facilitate P2P independent energy trading within microgrids. They designed two frameworks that used the smart contract feature as well as the continuous double auction and uniform-price double-sided auction mechanisms, respectively. Li and Zhang (Li & Zhang, 2021) considered the heat and electricity distribution at the same time with a P2P energy sharing mechanism based on a double auction to achieve the coordination of smart home microgrids. An integrated model was developed based on a combination of model-driven optimization and a data-driven prediction that could analyze data from past transactions.

As above-said, many articles have been published on P2P energy trading. Table 1 compares some articles to identify research gaps in such a domain better:

- Modeling approaches: Different types of decision variables, including continuous, binary, and integer variables, can be used to determine different mathematical models. These modeling approaches can be categorized as Linear Programming (LP), Mixed Integer Programming (MIP), Nonlinear Programming (NLP), Mixed-Integer linear programming (MILP), Mixed-Integer Nonlinear Programming (MINLP), Stochastic Linear Programming (SLP) as well as Game Theory.
- Objective Functions: In some studies, only one goal (Single) has been considered, while in others, several goals (Multi) have been considered.
- Solving methods: Exact method (E), and Simulation techniques (S).
- Research with consideration: Battery Storage (B), Uncertainty (U), Costs (C), Energy generation and consumption (G), and Pricing (P).
- The last two columns also list the main features as well as the case study of the article.

According to the literature review, there are limited studies on P2P energy trading under uncertainty and energy pricing, simultaneous; analysis of the performance of peers' battery and central battery storage, and other criteria such as the importance of time and cost of load shedding. Thus, this study presents a robust mathematical model to address the uncertainty of energy generation of DG and the demand of peers in a P2P grid in which energy is traded directly among peers on the

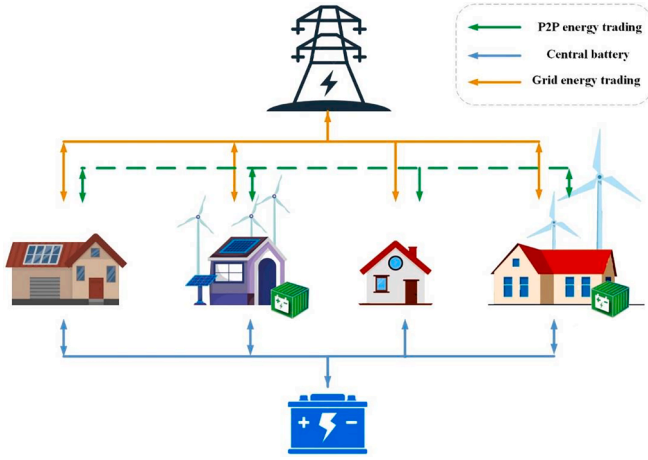


Fig. 2. P2P energy trading model.

blockchain platform in order to minimize costs and determine the amount of traded energy. Moreover, the pricing model is presented to assess energy prices, and it is solved by the Nash bargaining theory to boost the profit of peers and encourage them to participate in P2P.

3. Problem statement

The P2P energy trade network is the issue under discussion. As seen in Fig. 2, each home may be fitted with a variety of DG, including solar panels and wind turbines, as well as batteries that not only supply their demand but also sell any excess to the grid, a central battery, or other neighbors. A home without any DG is merely an energy consumer; however, homes with the equipment also play the producing role in addition to that of an energy consumer. The robust optimization approach used in this work helps to address the uncertainties associated with renewable energy generation because it is a weather-dependent process. The objective function is to minimize the cost of each home, which includes the cost of buying and selling electricity, the cost of battery depreciation, and the cost of a power outage.

In this case, it is assumed that houses can participate in the P2P energy trade, and their costs are compared in two modes of P2P and its absence. The next important issue is determining the price of energy in a P2P to maximize the profits of all peers, for which the Nash bargaining method is used. The pricing problem is solved based on the results of the P2P energy trading model.

In the following, parameters and decision variables are used to formulate the P2P energy trading model as well as pricing model:

Indices:	
i, j	Peers (Houses and central battery storage)
t	Hours t in time horizon
Parameters:	
$Pb_i^{\max}, Ps_i^{\max}$	Upper bounds of power that peer i purchase or sell to the grid
$Pch_i^{\max}, Pdc_i^{\max}$	Upper bounds of battery power that battery storage i can be charged or discharged (kW)
$nch_i, ndch_i$	Efficiency of battery storage i when charging and discharging
$Es_i^{\min}, Es_i^{\max}$	Upper/lower bounds of storage level in battery storage i (kWh)
PD_i^t	The projected demand for peer i at hour t
Pdg_i^t	The predicted generation capacity of the renewable peer i at hour t
Cs^t	The selling price of electricity to the grid at hour t
Cb^t	The purchase price of electricity from the grid at the hour t
Cs	Cost of storage depreciation at the time of charging and discharging
Cl_{si}	Cost of load shedding for peer i
Decision variables:	
Ub_i^t	If peer i buys electricity from the grid at the hour t , it is equal 1; otherwise 0
Pb_i^t	The amount of energy purchased from the grid for peer i at hour t
Ps_i^t	The amount of energy sold to the grid for peer i at hour t
uch_i^t	If battery storage i charge at hour t , it is equal 1; otherwise 0
$udch_i^t$	If battery storage i discharge at hour t , it is equal 1; otherwise 0

(continued on next column)

(continued)

Pch_i^t	The amount of charge of battery storage i at hour t
Pdc_i^t	The amount of discharge of battery storage i at hour t
Es_i^t	State of charge of battery i at hour t
Pe_{ij}^t	P2P electricity trades between peer i and peer j at hour t
$PDcur_i^t$	Load shedding rate for peer i at hour t
$Pdem_i^t$	The satisfied demand of peer i at hour t
$Pgpur_i^t$	The amount of generative power of DG taking into account the fluctuations of generation and consumption of peer i at hour t

Based on this problem description and defined notifications, the P2P energy trading problem under uncertainty and pricing problem is modeled and solved.

4. P2P energy trading model

The P2P energy trading problem is modeled, and then based on the results of this mathematical model, the pricing problem is formulated. Firstly, the mathematical model's constraints and the objective function are described.

The constraints regarding the restrictions on buying and selling energy with the grid for each house are stated as Equations (1) and (2):

$$0 \leq Pb_i^t \leq Ub_i^t \cdot Pb_i^{\max} \forall i, t \quad (1)$$

$$0 \leq Ps_i^t \leq (1 - Ub_i^t) Ps_i^{\max} \forall i, t \quad (2)$$

The restrictions on the battery storage include charging and discharging constraints, as shown in Equations (3) and (4). As simultaneous charging and discharging are impossible, the restriction is shown in Equation (5). Additionally, state of charge (SoC) updating that deepens on time and battery efficiency is described in Equation (6). The final constraint related to BESS is their SoC which is formulated in Equation (7).

$$0 \leq Pch_i^t \leq uch_i^t \cdot Pch_i^{\max} \forall i, t \quad (3)$$

$$0 \leq Pdc_i^t \leq udch_i^t \cdot Pdc_i^{\max} \forall i, t \quad (4)$$

$$uch_i^t + udch_i^t \leq 1 \forall i, t \quad (5)$$

$$Es_i^t = Es_i^{t-1} + nch_i \cdot Pch_i^t \cdot \Delta t - Pdc_i^t \cdot \Delta t / ndch_i \forall i, t \quad (6)$$

$$Es_i^{\min} \leq Es_i^t \leq Es_i^{\max} \forall i, t \quad (7)$$

The next constraint is related to the energy balance for each peer without considering P2P energy trading is described in Equation (8). In the case that the peers are not allowed to buy and sell energy simultaneously, the energy balance constraint includes demand (PD_i^t), amount of power load shedding ($PDcur_i^t$), amount of charge (Pch_i^t) and discharge (Pdc_i^t), the amount of energy buying (Pb_i^t) and selling (Ps_i^t) to the grid, and the amount of energy generation capacity for each peer (Pdg_i^t).

$$PD_i^t - PDcur_i^t + Pch_i^t - Pdc_i^t + Ps_i^t - Pb_i^t - Pdg_i^t = 0 \forall i, t \quad (8)$$

On the other hand, considering P2P energy trading, the energy balance constraint is shown in Equation (9).

$$PD_i^t - PDcur_i^t + Pch_i^t - Pdc_i^t + Ps_i^t - Pb_i^t - Pdg_i^t + \sum_j Pe_{ij}^t = 0 \forall i, j, t \quad (9)$$

The power selling by peer i to peer j must be equal to the power purchasing by peer j from peer i ; this constraint is modeled as Equation (10).

$$Pe_{ij}^t + Pe_{ji}^t = 0 \forall i, j, t \quad (10)$$

As mentioned, regardless of the P2P, the cost of each house encompasses the cost of buying and selling energy with the grid, battery depreciation cost, and the cost of load shedding. These costs are calculated by Equations (11) to (13), respectively.

$$C_i^{trade} = \sum_t (Pb_i^t \cdot Cb^t - Ps_i^t \cdot Cs^t) \quad (11)$$

$$C_i^{ES} = C_s \sum_t (Pch_i^t \cdot \Delta t + Pdch_i^t \cdot \Delta t) \quad (12)$$

$$C_i^{LS} = Cls_i \sum_t (PDcur_i^t \cdot \Delta t) \quad (13)$$

Finally, the total cost of each house is calculated as Equation (14).

$$C_i = C_i^{trade} + C_i^{ES} + C_i^{LS} \quad (14)$$

Therefore, the studied problem includes the objective function (14) with constraints (1) to (13). It should be mentioned that the cost of energy trading in P2P will be added to these costs after the optimal energy price of P2P trading is calculated in Section 5.

4.1. Robust optimization formulation

In this study, the robust optimization approach that was developed by Bertsimas and Sim (Bertsimas & Sim, 2004) is used to tackle the uncertainty of the proposed mathematical model. The cardinality-constrained robust optimization approach allows the decision maker to consider a certain degree of uncertainty for the electricity generation of renewable resources and the amount of demand of peers that are modeled coordinately together.

The UN_L and UN_R are assumed to be the percentage of changes in demand and the electricity generation, respectively. Moreover, considering the PD_i^t as the average demand and the $Pdem_i^t$ as the satisfied demand. Equations (15) and (16) apply to the demand:

$$Pdem_i^t \leq PD_i^t \cdot (1 + UN_L) \Rightarrow \Delta Pdem_i^t = Pdem_i^t - PD_i^t \leq PD_i^t \cdot UN_L \quad (15)$$

$$Pdem_i^t \geq PD_i^t \cdot (1 - UN_L) \Rightarrow \Delta Pdem_i^t = Pdem_i^t - PD_i^t \geq -PD_i^t \cdot UN_L \quad (16)$$

Furthermore, considering the Pdg_i^t as the average predicted power generation for renewables resources, and the Pg_i^t as the satisfied power from resources, the Equations (17) and (18) apply to the electricity generation:

$$Pg_i^t \leq Pdg_i^t \cdot (1 + UN_R) \quad (17)$$

$$Pg_i^t \geq Pdg_i^t \cdot (1 - UN_R) \quad (18)$$

Based on Equations (15) to (18), Equations (19) and (20) are formulated in order to consider the simultaneity of demand and energy generation uncertainty.

$$Pgpur_i^t = Pg_i^t - \Delta Pdem_i^t \leq Pdg_i^t \cdot (1 + UN_R) + PD_i^t \cdot UN_L \quad (19)$$

$$Pgpur_i^t = Pg_i^t - \Delta Pdem_i^t \geq Pdg_i^t \cdot (1 - UN_R) - PD_i^t \cdot UN_L \quad (20)$$

Clearly, the uncertainty of electricity generation and the uncertainty of demand can be modeled simultaneously. To this end, we consider the variable $Pgpur$ as the amount of electricity generation by taking into account generation and consumption fluctuations. With this assumption, the equation related to the energy balance of each peer changes as Equation (21).

$$Pgpur_i^t \geq PD_i^t - PDcur_i^t + Pch_i^t - Pdch_i^t + Ps_i^t - Pb_i^t + \sum_j Pe_{ij}^t \quad (21)$$

In the following, according to Equations (19) and (20), and the assuming δ_1 and δ_2 as dual variables, we obtain the Equations (22) and (23) as below:

$$Pgpur_i^t \leq Pdg_i^t \cdot (1 + UN_R) + PD_i^t \cdot UN_L \quad : \delta_1^t \quad (22)$$

$$-Pgpur_i^t \leq -Pdg_i^t \cdot (1 - UN_R) + PD_i^t \cdot UN_L \quad : \delta_2^t \quad (23)$$

Considering the worst-case scenario modeling of uncertainty for Equation (21), the Equation (24) is formulated:

$$\max(-Pgpur_i^t) \leq -PD_i^t + PDcur_i^t - Pch_i^t + Pdch_i^t - Ps_i^t + Pb_i^t - \sum_j Pe_{ij}^t \quad (24)$$

Using the duality theorem for uncertainty, the following model is obtained:

$$\min \delta_1^t \cdot (Pdg_i^t \cdot (1 + UN_R) + PD_i^t \cdot UN_L) + \delta_2^t \cdot (-Pdg_i^t \cdot (1 - UN_R) + PD_i^t \cdot UN_L) \quad (25)$$

$$\delta_1^t - \delta_2^t \leq -1 \quad (26)$$

$$\delta_1^t \geq 0, \quad \delta_2^t \geq 0; \quad (27)$$

Therefore, considering the uncertainty, the imbalance constraint is replaced as Equation (28):

$$\delta_1^t \cdot (Pdg_i^t \cdot (1 + UN_R) + PD_i^t \cdot UN_L) + \delta_2^t \cdot (-Pdg_i^t \cdot (1 - UN_R) + PD_i^t \cdot UN_L) \leq -PD_i^t + PDcur_i^t - Pch_i^t + Pdch_i^t - Ps_i^t + Pb_i^t - \sum_j Pe_{ij}^t \quad (28)$$

Finally, considering the uncertainty, the mathematic model is defined as follows:

- The objective function of Equation (14)
- The constraints (1) to (7), (10) to (13), and constraints (26) to (28)

The problem-solving method is described in the next section.

4.2. P2P energy trading problem solving method

The described problem involves the minimization of Equation (14), which includes the cost of each peer. The minimization needs to be solved by considering the cost function of all peers simultaneously to obtain the best answer for this relation. However, we cannot do that for our problem. Firstly, we assume that there is no central unit for monitoring trading, and trades are made based on a blockchain platform which means peers connect with each other directly. Secondly, most peers are reluctant to share their information with other units for security reasons. Therefore, cost minimization has to be solved using a distributed algorithm.

The Alternating Direction Method of Multiplier (ADMM) is known as one of the best methods for solving problems in a distributed algorithm. In this method, the problem is solved in two levels. It should be noted that the energy trading problem among peers is initially solved; then, the electricity price of P2P trading is determined based on the energy trading and profit in such a way that the peers would benefit from the profit according to their participation rate in the P2P energy trading.

First, the energy trading problem among peers is solved using ADMM. According to the ADMM algorithm, this problem is split into the following two parts:

1. Lower level optimization (LLO) problem

At this level, each peer independently solves their scheduling problem for the next 24 h. It is assumed that the peers do this task and record its results, including the amount of electricity exchanged with other peers, on the blockchain platform, which all other peers can see.

2. Upper-Level optimization (ULO) problem

At this level, based on the result obtained from the previous level, i. e., the amount of exchange electricity of each peer with other peers, the amount of exchange would be optimized in such a way to optimize the amount of power exchange while being close to the desired power of the peers.

The LLO and ULO problems are solved repeatedly until the answers of the two levels for Pe_{ij}^t become the same, which process is described below. A covariate is initially assumed as $\widehat{Pe}$ for Pe_{ij}^t such that the main

problem is converted into a two-part problem according to the ADMM process. The $\widehat{Pe}$ covariate must satisfy the following constraints:

$$Pe_{ij}^t = Pe_{ji}^t : \lambda_{ij}^t \quad (29)$$

$$Pe_{ij}^t + Pe_{ji}^t = 0 \quad (30)$$

According to Equation (29), λ_{ij}^t is considered as the dual of this equation. Based on the ADMM algorithm, the Lagrangian function of the problem needs to be modeled, which is done as follows.

$$L_1 = \sum_i C_i + \sum_j \sum_i \left(\frac{\rho_1}{2} (Pe_{ij}^t - Pe_{ij}^t)^2 + \lambda_{ij}^t (Pe_{ij}^t - Pe_{ij}^t) \right) \quad (31)$$

In this equation ρ_1 is a parameter that acts as a penalty for violating the constraint (29).

Problem-solving with the ADMM approach involves repetitive solving of the LLO and ULO problems. As mentioned, the LLO problem is used to solve the local optimization problem for each peer. The value of the dual variable λ_{ij}^t and the covariate value Pe_{ij}^t are needed to solve this problem. The goal of the ULO problem is also to update the value of the dual variable λ_{ij}^t and the value of the covariate Pe_{ij}^t , which is solved based on the value of the Pe_{ij}^t obtained from the LLO problem.

In the problem solving process, assuming to be in iteration k , the i -th peer solves its optimization problem based on the objective function (32) by considering the constraints (1) to (7), (11) to (14), and (26) to (28).

$$LLO : \min C_i + \sum_j \sum_i \left(\frac{\rho_1}{2} (Pe_{ij}^t - Pe_{ij}^t)^2 - \lambda_{ij}^t . Pe_{ij}^t \right) \quad (32)$$

st. (1) – (7), (11) – (14), (26) – (28)

Once the problem (32) is solved for all peers, the value of $Pe_{ij}^t(k+1)$ will be obtained. Continuing the process, this value is used to update the values of the dual variable λ_{ij}^t and the covariate value $\widehat{Pe}_{ij}^t$ in the ULO problem. The ULO is defined as follows.

$$ULO : \min \sum_i \sum_j \sum_i \left(\frac{\rho_1}{2} (Pe_{ij}^t - Pe_{ij}^t(k+1))^2 + \lambda_{ij}^t . Pe_{ij}^t \right) \quad (33)$$

st. (30)

By solving the problem (33), which can be done by all the peers individually on the blockchain platform, the value of $Pe_{ij}^t(k+1)$ is obtained. Accordingly, the dual variable λ_{ij}^t is updated based on the ADMM algorithm as follows.

$$\lambda_{ij}^t(k+1) = \lambda_{ij}^t(k) + \rho_1 (Pe_{ij}^t(k+1) - Pe_{ij}^t(k+1)) \quad (34)$$

Based on the explanations provided, the peers cost optimization problem can be solved by considering the P2P in the following steps:

1. Defining the initial values, including the repeat counter ($k = 0$), penalty coefficient value ($\rho_1 = 1$), the dual variable value ($\lambda_{ij}^t(0) = 0$), and the covariate value ($Pe_{ij}^t(0) = 0$).
2. Solving the LLO problem based on the values of the dual variable and the covariate and obtaining the value of Pe_{ij}^t for all peers.
3. Solving the ULO problem based on Pe_{ij}^t values and obtaining the value of Pe_{ij}^t .
4. Updating dual variable λ_{ij}^t .
5. If $\|Pe_{ij}^t(k+1) - Pe_{ij}^t(k+1)\| \leq Th_1$, the problem is converged, otherwise, we have $k = k + 1$, and we go to step 2.

The above five steps are solved repeatedly, and once the convergence condition is met, the answer to the problem is determined, which in-

cludes the amount of power purchased and sold by each peer with the grid (Pb_i^t, Ps_i^t), charge and discharge battery storage for each peer ($Pch_i^t, Pdch_i^t$), and power exchanged among peers and the central battery storage (Pe_{ij}^t). In step 5, the convergence condition is that the second norm of the difference in the value of Pe_{ij}^t obtained at the upper and lower levels is lower than the threshold value (Th_1).

After solving this problem, the amount of power exchanged and the total profit will be obtained from the P2P. Accordingly, the price of this electricity trading and the payment amount from each peer to another peer and the central battery storage is determined in the next section.

5. Pricing model

After determining the amount of energy traded among peers and the grid as well as the central battery storage, the energy price is determined in order to maximize the profits of peers in this sector. The Nash bargaining game (Binmore, Rubinstein, & Wolinsky, 1986) is used to solve the pricing problem; in this situation, using Nash bargaining is one of the best bargaining theories (Nash, 1951; Roth and Kagel, 1995).

The bargaining game that is related to game theory considers the number of peers who have a set of feasible outcomes, and they are specified by unanimous agreement for them. As at least two of the peers have a different outcome that is most preferable, there is a need for bargaining over which outcome should be agreed upon. Since the agreement is required, each peer has the ability to veto any outcome. According to this method, the problem is balanced when the multiplication of the peers' profits is maximized, and the peers' profits would be proportional to their participation in the P2P. It should be noted that in this case, the peers in the problem make decisions simultaneously.

Accordingly, the pricing problem and sharing of profit resulting from the P2P are defined as follows. The problem's objective function and constraints are described below.

$$\max \prod_i (\Delta C(i) - C_{p2p}(i)) \quad (35)$$

s.t.

$$\Delta C(i) = C_{\min}(i) - C^*(i) \forall i \quad (36)$$

$$C^*(i) + C_{p2p}(i) \leq C_{\min}(i) \forall i \quad (37)$$

$$C_{p2p}(i) = \sum_j \sum_i pay(i, j, t) \forall i \quad (38)$$

$$pay(i, j, t) = -Pe(i, j, t) . Pr_{p2p}(i, j, t) \forall i, j, t \quad (39)$$

$$Pr_{p2p}^{\min} \leq Pr_{p2p}(i, j, t) \leq Pr_{p2p}^{\max} \forall i, j, t \quad (40)$$

Equation (35) is the objective function of the problem, which involves maximizing the profits of all peers based on their participation rate in the P2P. In Equation (36), $\Delta C(i)$ is the amount of change in the cost of peer i due to its participation in the P2P, while $C_{\min}$ is the amount of the cost of each peer without considering P2P that is obtained from the previous section. $C^*(i)$ is the amount of the cost of each peer regarding the P2P that is obtained from the previous section. Moreover, $Pe(i, j, t)$ is the amount of power traded between peer i and peer j at hour t that is obtained from the P2P energy trading model; furthermore, $Pr_{p2p}^{\min}$ and $Pr_{p2p}^{\max}$ describe minimum and maximum energy prices in P2P.

In Equation (37), $C_{p2p}(i)$ is the amount of cost that peer i has paid to other peers for purchasing power in the P2P or the amount of money received from the sale of power to other peers. Equation (37) ensures that a peer's participation in the P2P would not increase its cost. In Equation (38), $pay(i, j, t)$ is the payment amount of peer i to peer j at hour t , and in this equation, the cost or revenue from participating in the P2P is calculated for all peers. In Equation (39), the amount paid by the peers to each other is calculated based on the predetermined price (Pr_{p2p}), and

Table 2

The information of houses.

Type of power source	House 1	House 2	House 3	House 4	Central battery storage
Wind (kW)	40	–	40	–	–
PV (kW)	–	–	60	60	–
Maximum power interchangeable with the grid (kW)	50	50	50	50	–
Maximum charging power (kW)	10	–	10	–	40
Maximum battery capacity (kWh)	30	–	30	–	100

the amount of power trading among the peers. Equation (40) keeps the value of the two-way trading within a certain range.

5.1. Pricing problem solving method

Similar to the previous section, the pricing problem is solved by the ADMM algorithm. Therefore, the covariate $\widehat{pay}$ is introduced to satisfy Equations (41) and (42).

$$\widehat{pay}(i,j,t) = pay(i,j,t) : \mu(i,j,t) \quad (41)$$

$$\widehat{pay}(i,j,t) + \widehat{pay}(j,i,t) = 0 \quad (42)$$

Clearly, the $\mu(i,j,t)$ has been considered as the dual variable of Equation (41). According to ADMM, the objective function (35) will be converted to the Lagrangian function shown below following the logarithmic conversion.

$$L_2 = \sum_i \left[-\ln(\Delta C(i) - C_{p2p}(i)) + \sum_j \sum_t \left(\frac{\rho_2}{2} (\widehat{pay}_{ij}^t - pay_{ij}^t)^2 + \mu_{ij}^t (\widehat{pay}_{ij}^t - pay_{ij}^t) \right) \right] \quad (43)$$

In this Equation, ρ_2 is the penalty parameter for the constraint (41).

As in the previous section, the pricing problem is solved as a two-level problem by a repetition approach. The Lower-Level Pricing (LLP) problem is defined as Equation (44):

$$LLP : \min -\ln(\Delta C(i) - C_{p2p}(i)) + \sum_j \sum_t \left(\frac{\rho_2}{2} (\widehat{pay}_{ij}^t - pay_{ij}^t)^2 - \mu_{ij}^t \cdot pay_{ij}^t \right) \quad s.t \quad (36) - (40) \quad (44)$$

After solving the problem (44) by each peer, the value of $pay_{ij}^t(k+1)$ is obtained, by using which, the ULO problem will be solved to obtain the optimal values of $\widehat{pay}_{ij}^t$. The Upper-Level Pricing (ULP) problem is defined as Equation (45).

Table 3

The optimization results.

-	Cost with P2P (\$)	Cost without P2P (\$)	Profit (\$)
House 1	39.17	49.86	10.69
House 2	105.35	117.59	12.24
House 3	-86.92	-77.7	9.22
House 4	40.05	50.38	10.33
Central battery storage	-9.46	0	9.46
Total (\$)	88.19	140.13	51.94

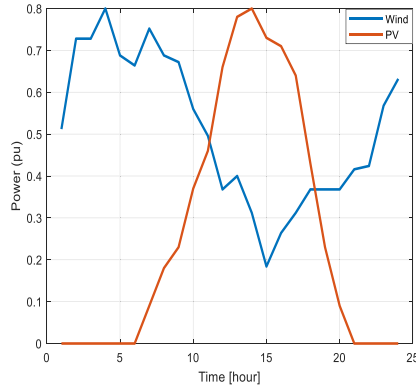
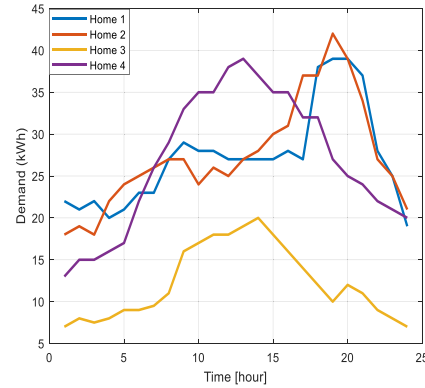
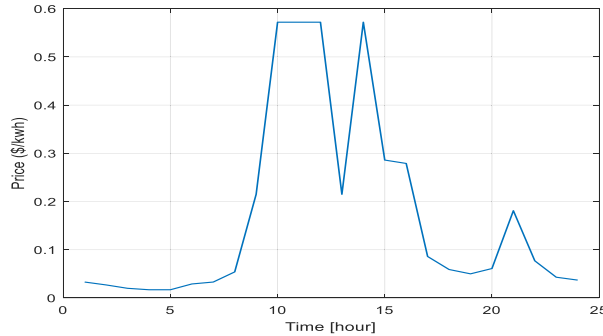
**(a)****(b)****(c)**

Fig. 3. The parameters which (a) the energy consumption pattern of houses, (b) the amount of normalized power predicted for the average wind and solar power, (c) Feed-in-tariff.

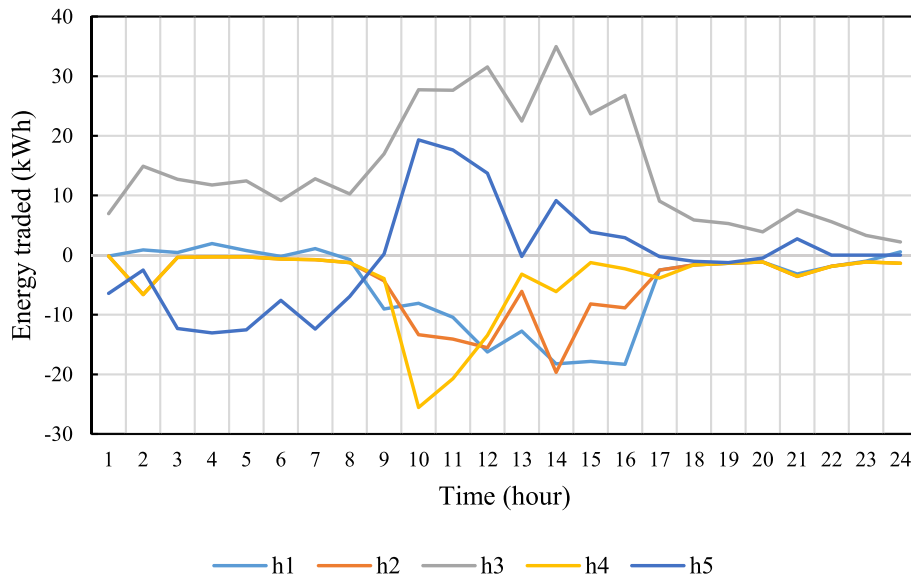


Fig. 4. The exchange power between houses.

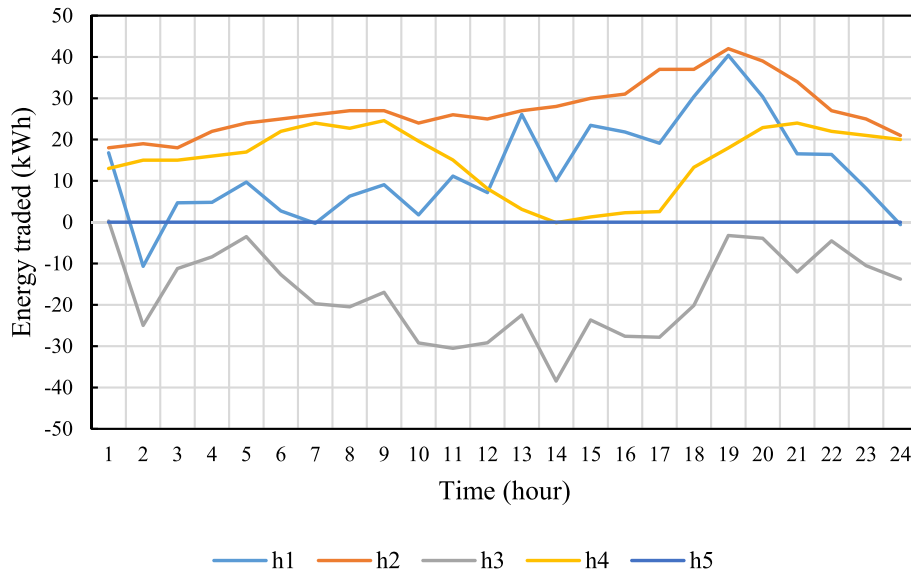


Fig. 5. The exchange power of houses with the grid without considering the P2P.

$$ULP: \min \sum_j \sum_t \left(\frac{\rho_2}{2} (\widehat{pay}_{ij}^t - pay_{ij}^t)^2 + \mu_{ij}^t \widehat{pay}_{ij}^t \right) \quad (45)$$

s.t (42)

By solving the ULP problem, the optimal value of $\widehat{pay}_{ij}^t(k+1)$ is obtained, and accordingly, the value of the dual variable μ_{ij}^t will be updated as Equation (46).

$$\mu_{ij}^t(k+1) = \mu_{ij}^t(k) + \rho_2 (\widehat{pay}_{ij}^t(k+1) - pay_{ij}^t(k+1)) \quad (46)$$

According to the explanations provided, the optimal pricing problem for the P2P can be solved according to the following steps:

1. Defining the initial values, including the repeat counter ($k = 0$), penalty coefficient value ($\rho_2=1$), the dual variable value ($\mu_{ij}^t(0) = 0$), and the covariate value ($\widehat{Pay}_{ij}^t(0) = 0$).
2. Solving the LLP problem based on the values of the dual variable and the covariate and obtaining the value of pay for all peers.
3. Solving the ULP problem based on pay values and obtaining the value of $\widehat{pay}_{ij}^t$.
4. Updating dual variable μ_{ij}^t .
5. If $\|\widehat{pay}_{ij}^t(k+1) - pay_{ij}^t(k+1)\| \leq Th_2$, the problem is converged, otherwise, we have $k = k + 1$, and we go to step 2.

By solving the optimal pricing problem, the amount paid or received by each peer from the other peers is determined, followed by specifying the price of bilateral trading.

6. Case study

In this section, we collected data from Iran to evaluate the proposed mathematical model. The data has been collected is electricity price, FiT, solar radiation, and wind speed. In this simulation, considered four

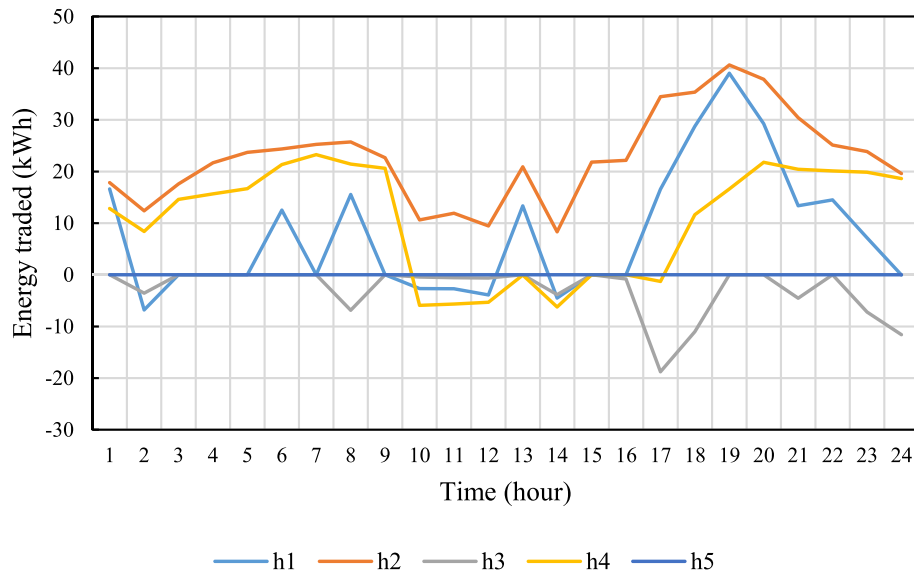


Fig. 6. The exchange power of houses with the grid with considering the P2P.

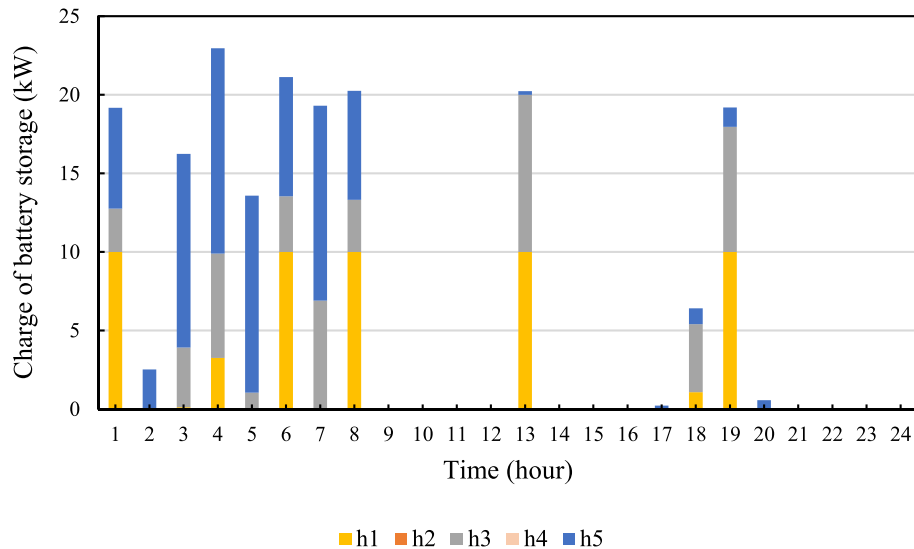


Fig. 7. The amount of electricity to charge batteries.

houses that are equipped with DG, such as PV and wind turbines, as well as battery storage which is shown in Table 2.

Other parameters are shown in Fig. 3; Fig. 3(a) shows the amount set for the energy consumption of houses, while Fig. 3(b) shows the amount considered for the average wind and solar power. Finally, Fig. 3(c) indicates FiT that the grid purchases the energy.

The FiT is considered lower than the purchase price to encourage houses to use renewable resources locally. For this reason, the selling price of energy to the grid is considered equal to 70% of the purchase price.

The optimization results are obtained by considering 15% uncertainty for renewable resources and 10% uncertainty for demand, as shown in Table 3.

According to Table 3, running P2P energy trading in the local grid can reduce costs and also increase profits for all houses. For example, in this case study, P2P can reduce the total cost of houses by \$51.94, which is highly significant.

The amount of electricity trading among houses is shown in Fig. 4, which in the positive values relate to selling electricity to others. As seen

in the figure, House 3, which generates significant electricity, sells the surplus power to other houses or the central battery storage at all hours. It is also seen that the central battery storage buys energy from House 3 during off-peak hours and sells it to Houses 1, 2, and 4 during peak hours.

Fig. 5 shows the amount of power that is traded between houses and the grid without considering P2P energy trading. Then, Fig. 6 describes how much power is traded between houses and the grid by applying P2P during the day.

It is clear, applying P2P affects energy trading, and P2P can significantly reduce exchange power with the grid. For instance, House 3, before applying P2P energy trading, sells its extra power to the grid, while after using P2P, the power that is sold to the grid is reduced to zero or much lower than the original state at most hours. Also, Houses 1 and 4 manage power to sell energy to the grid during peak hours by purchasing energy and charging the battery storage.

Various analyzes can also be performed on the peers' batteries and the central battery. Fig. 7 shows the amount of electricity that charges the batteries, and Fig. 8 shows the amount of electricity discharged from

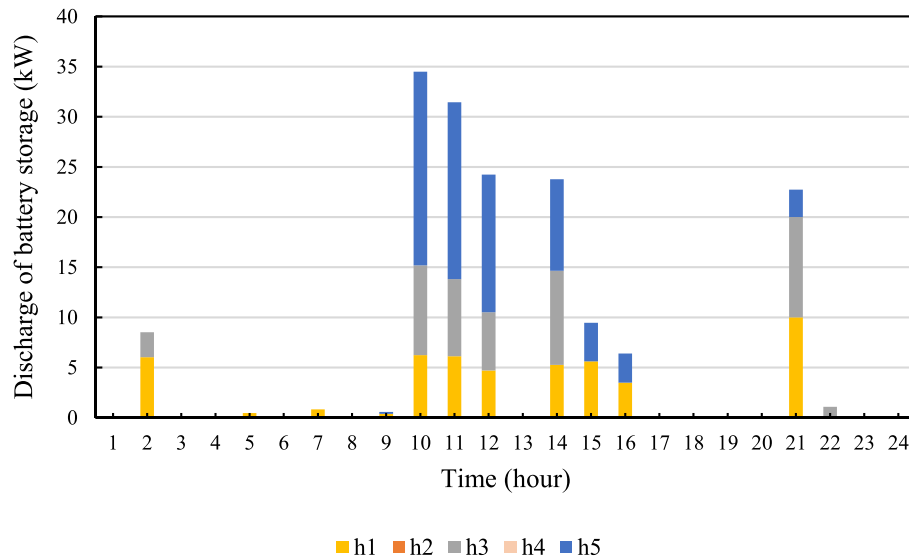


Fig. 8. The amount of electricity discharged from batteries.

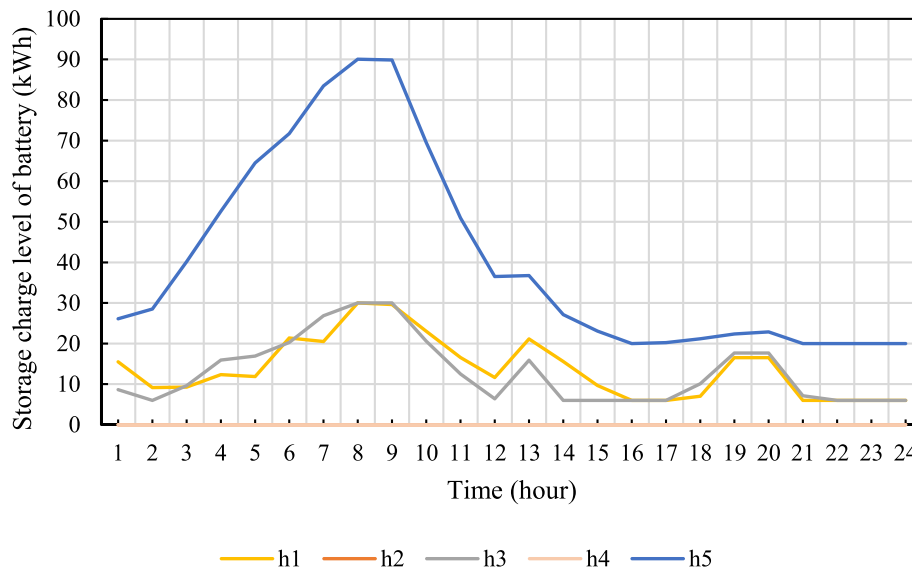


Fig. 9. Storage charge level of batteries.

the batteries. As shown in Fig. 7, during off-peak hours, excess energy is charged in the batteries to be used during peak hours. These two figures show that the central battery is charged during off-peak hours, and the stored energy is used in the middle of the day when consumption is high.

Fig. 9 shows the level of electricity stored in batteries. The central battery stores more energy due to its high capacity, and as it was clear from Fig. 7, this storage is done gradually during low consumption hours, and according to Fig. 8, it is discharged during peak consumption hours. To prevent the batteries' depreciation, as it turns out, the batteries do not drain completely and have some energy in them.

The next significant output of the mathematical model is to determine the price per hour for each house. According to Nash's bargaining theory, the price per house per hour in P2P is calculated to maximize profits. These energy prices are shown separately for each house in Fig. 10.

Fig. 10(a) shows the energy price in P2P for House 1 from other houses. These prices for Houses 2, 3, and 4 as well as the central battery are shown in Fig. 10(b), (c), and (d), respectively.

It is clear that the energy price of House 3 is the highest at most hours

because this house is the most equipped and has the highest energy generation. Additionally, energy prices reach their highest during peak hours, and energy prices are low during low hours. The amount paid by each house in the P2P is obtained by multiplying these prices in each hour by the amount of energy exchanged.

7. Sensitivity analysis

To better understand the proposed model performance, sensitivity analysis of the effective parameters, degree of uncertainty for the electricity generation of renewable resources (UN_R) and the amount of demand of peers (UN_L), and maximum battery capacity ($E_S^{\max}$), are analyzed. The changing of decision variables are investigated based on the different parameters in the model.

7.1. Degree of uncertainty

In this section, the effect of the change in the demand of peers as well as the amount of energy generation by DG on peers' profits is examined.

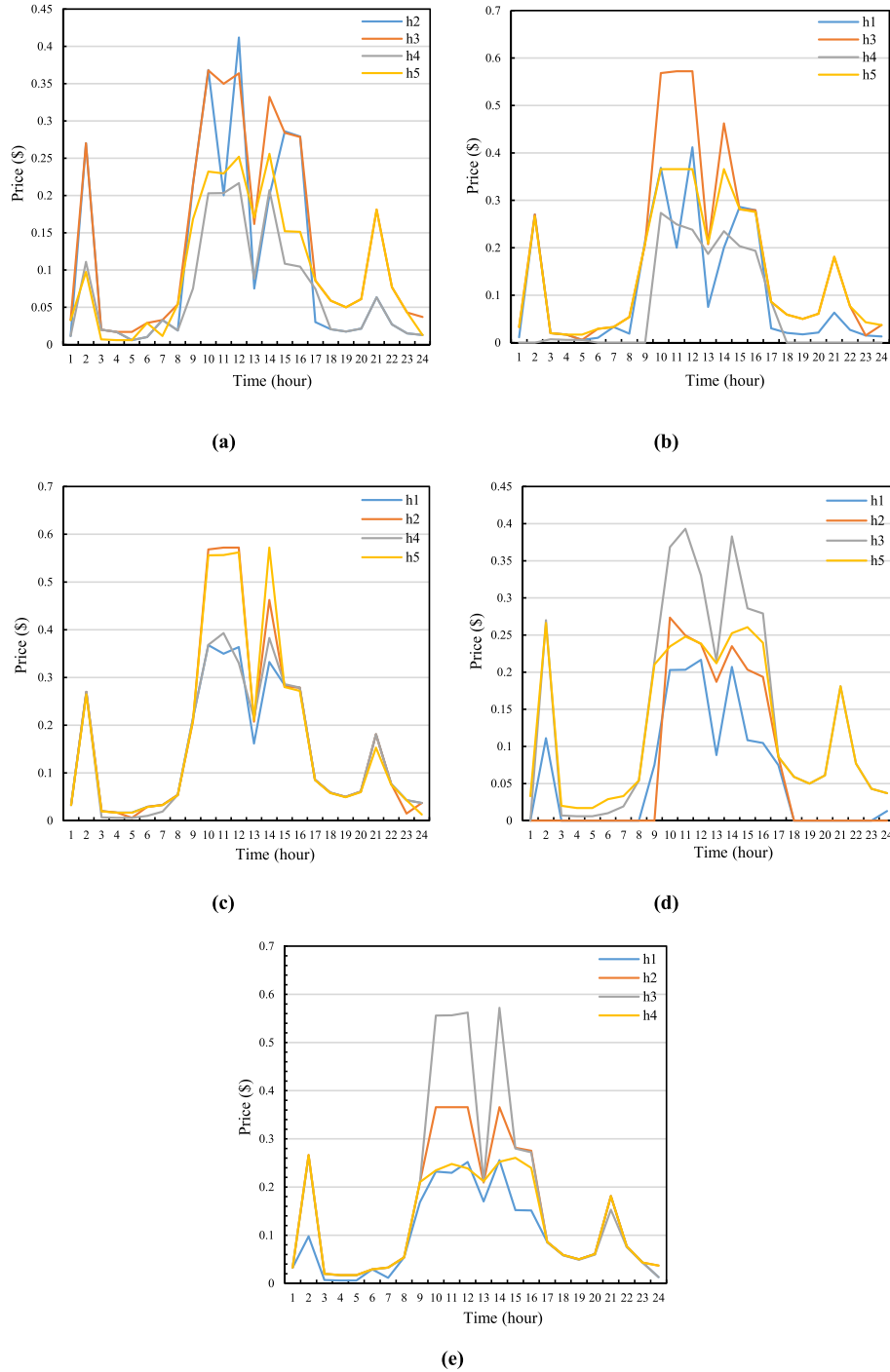


Fig. 10. Price of power in P2P for (a) House 1, (b) House 2, (c) House 3, (d) House 4, (e) central battery.

In order to investigate the effect of fluctuations in energy generation on the total profit of peers, the degree of demand uncertainty is kept constant (10%), but the amount of energy generation uncertainty is changed. These outputs are shown in Fig. 11(b). Moreover, in order to investigate the effect of demand fluctuations on the total profit of peers, with the constant amount of energy generation uncertainty (15%), the uncertainty in peers' demand is changed, and the outputs of these effects are shown in Fig. 11(a).

As shown in Fig. 11, fluctuations in energy generation have a greater effect on total profit than fluctuations in demand. It is also found that the greater the uncertainty of both parameters, the lower the overall profit.

7.2. Battery capacity

The effect of the battery capacity of houses and the central battery on the amount of energy storage is examined in this section.

7.2.1. Central battery capacity

First of all, the effect of changes in central battery capacity on SoC of the homes' batteries is investigated. What is clear is that if the capacity of the central battery is increased, more energy is stored in this battery, as shown in Fig. 12(c). But another interesting point is that the capacity of the central battery changes SoC of the homes' batteries at different hours. For example, Fig. 12(a) shows the SoC of the battery of House 1,

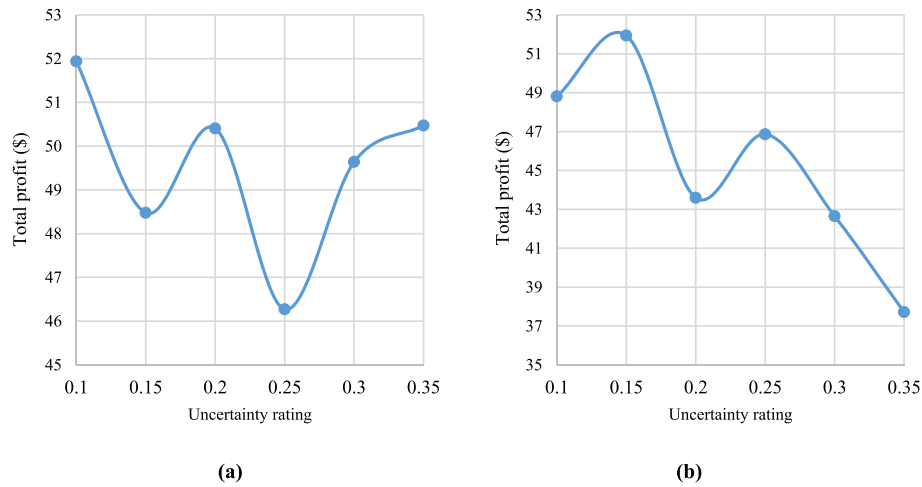


Fig. 11. Effect of demand and energy generation fluctuations on total profits; (a) percentage of demand uncertainty, (b) percentage of energy production uncertainty.

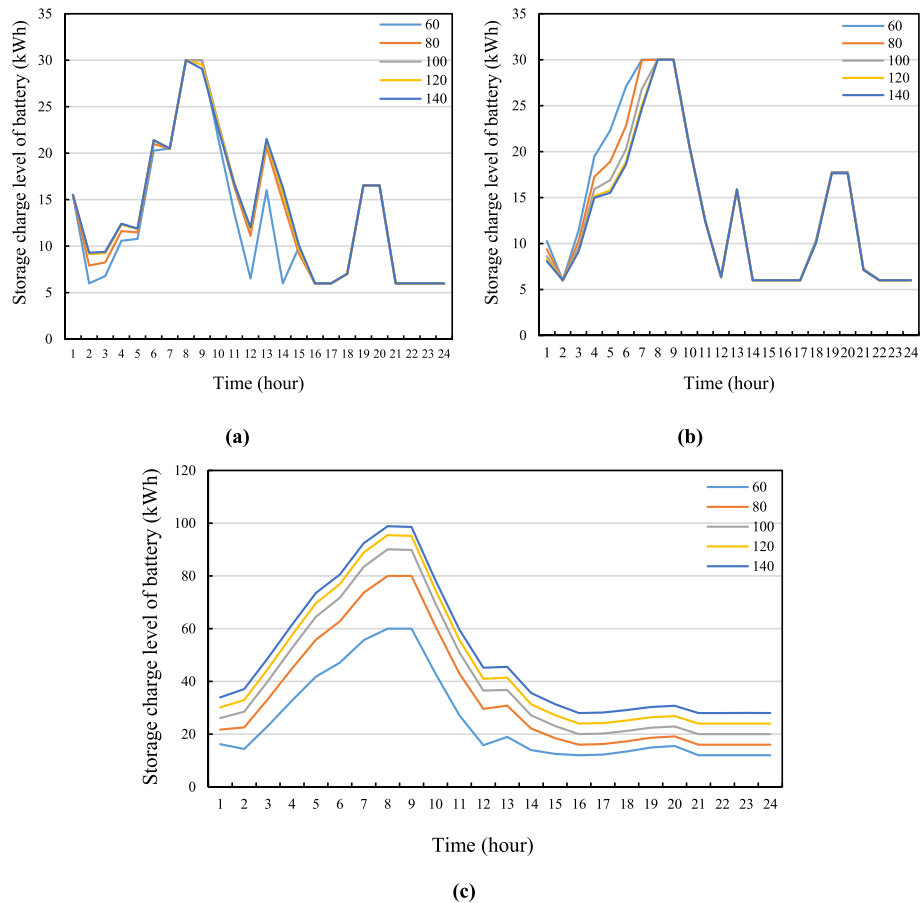


Fig. 12. Effect of central battery capacity on storage charge level of batteries; (a) House 1, (b) House 3, (c) central battery storage.

which shows that the low capacity of the central battery causes less energy to be stored in the battery of House 1 in some hours. This is because House 1 either shares more power in the P2P or uses power for its own consumption to compensate for the low capacity of the central battery. It should be noted that increasing the capacity of the central battery after a certain level has no effects.

On the other hand, Fig. 12(b) shows the House 3 battery has an entirely different behavior, in that by reducing the capacity of the central battery, the House 3 battery stores more energy.

7.2.2. Houses batter capacity

Like the capacity of a central battery, when the battery capacity of homes is increased, each of them stores a greater amount of energy. In fact, there is a direct relationship between capacity and the amount of storage in the homes. However, the effect of home batteries capacity on the amount of central battery storage is not direct but shows different behaviors at different hours, the results of which are shown in Fig. 13.

It is clear that when home batteries have the highest capacity, the central battery stores the least energy. However, this is not the case with

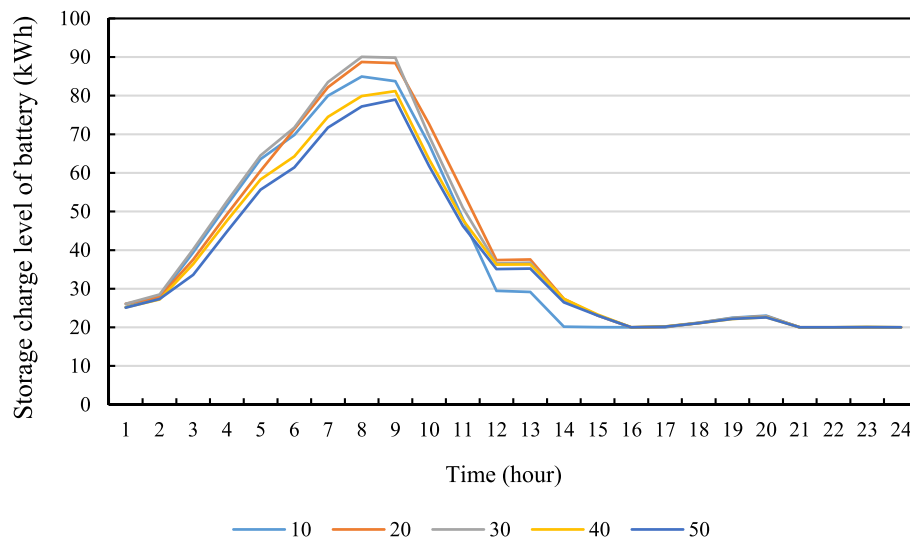


Fig. 13. Effect of houses battery capacity on storage charge level of central battery.

the lowest capacity of home batteries, and the most central storage is when home batteries have a medium capacity.

8. Conclusions

In this study, a mathematical model for P2P energy trading was introduced, including a central battery in the local grid, various DG, and batteries. All trades were conducted on the blockchain platform without the need for a supervisor. In order for all peers to maximize their profits, this study offered the pricing model to estimate the price of energy exchanged in the P2P energy trading using the Nash bargaining strategy. The suggested model was put into practice in a simulation framework using data from Iran, including average customer demand, wind speed, sunlight intensity, and power costs, and four houses with various equipment. The amount of demand of peers, as well as the amount of energy generation were assumed to be uncertain due to human activities and weather changes, which was tackled by cardinality-constrained robust optimization approach. The mathematical model was solved by ADMM to ensure that optimal solutions were obtained.

The created insights enabled us to examine the complex interactions between peers in P2P energy trading under two scenarios of whether or not to participate in P2P trading, which turned out to be beneficial P2P use for all peers; hence, policymakers and consumers are encouraged to use renewable energy and P2P energy trade. Saving energy at home depends on many factors, including the type of DG and the battery. This model showed that the maximum profit could be achieved by customers equipped with PV and wind turbines and battery storage. Moreover, the profits were much higher for all peers in the presence of the P2P energy trade.

In future research, smart contracts can be examined to determine the type of communication and increase business security. Furthermore, looking at the sustainability of the P2P energy trading and examining its social and environmental dimensions can be practical and reasonable incentives to make more use of renewable energy. Moreover, considering other components such as electric vehicles in this energy trading can be considered to achieve the comprehensiveness of the P2P energy trading.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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